



الهيئة الاتحادية للهوية
والجنسية والجمارك وأمن المنافذ
FEDERAL AUTHORITY FOR IDENTITY,
CITIZENSHIP, CUSTOMS & PORT SECURITY

ID Card Toolkit Programmer's Reference

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Revision History

Version	Date	Description of Changes
1.0	23 Feb 2018	Release Version 1.0.0.
1.1	27 May 2018	Release Version 1.0.5.
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1.9	20 Dec 2023	Release Version 3.0.0.
2.0	Mar 2026	Release Version 3.0.5. Added error codes, AdES parameters, PKCS#11, face authentication.

Abbreviations

Abbreviation	Description
API	Application Programming Interface
CAdES	CMS Advanced Electronic Signatures
ICP	Federal Authority for Identity, Citizenship, Customs & Port Security
JWS	Java Web Start
NFC	Near Field Communication
OTG	USB On-The-Go
PAdES	PDF Advanced Electronic Signatures
PCSC	Personal Computer Smart Card
PKCS#11	Public-Key Cryptography Standards #11
PKI	Public Key Infrastructure
REST	Representational State Transfer
SDK	Software Development Kit
SP	Service Provider
USB	Universal Serial Bus
VG	Validation Gateway, provided by ICP
VPN	Virtual Private Network
XAdES	XML Advanced Electronic Signatures

1. Introduction

The Federal Authority for Identity, Citizenship, Customs & Port Security (ICP) developed the ID Card Toolkit to address the requirements of service providers (SP) to integrate, into their business applications: Identification, Authentication, Digital Signature and Non-repudiation services, around the capabilities of the Emirates ID Card. The Toolkit is comprised of a number of Software Development Kits (SDK) supporting different programming languages and platforms. The supported programming languages include: C/C++, Java, Objective-C, Swift and C#.

Programming language specific developer guides and samples are provided as part of each SDK to support developers with Toolkit integration.

The Toolkit also provides a migration path for SPs that have existing web applications using ICP's Validation Gateway (VG) Applet and ActiveX components. With minimal changes an existing web application can be upgraded to the new Toolkit.

1.1. Purpose

This document provides reference information required by the Toolkit programmers irrespective of the programming languages and environments used.

2. Emirates ID Card Data Elements

This section lists the various data elements present on the Emirates ID Card.

2.1. Non-Modifiable Data

#	Attribute Name	Description	Remarks
1	IdNumber	Unique Identification Number for the card holder	Format: 784YYYYnnnnnnnC YYYY – Year of Birth nnnnnnn – unique number C – Check Digit
2	CardNumber	Unique Card Number of the card holder	Unique Card Number
3	IDType	ID Type	String values: "ID" – Identity Cards for Citizens "IR" – Identity Cards for Residents "IL" – Identity Cards for GCC
4	IssueDate	ID Card Issue Date	Format: DD/MM/YYYY
5	ExpiryDate	ID Card Expiry Date	Format: DD/MM/YYYY
6	TitleArabic	Title in Arabic	Refer to Toolkit References
7	FullNameArabic	Full Name in Arabic	Citizens: First, Father, Grandfather, Grand Grandfather, Family, Tribe, Clan Residents: First, Second, Third, Last, Family, Tribe, Clan
8	TitleEnglish	Title in English	Refer to Toolkit References
9	FullNameEnglish	Full Name in English	Same composition as FullNameArabic
10	Gender	Gender	M – Male, F – Female
11	NationalityArabic	Nationality in Arabic	Refer to Toolkit References
12	NationalityEnglish	Nationality in English	Refer to Toolkit References
13	NationalityCode	Nationality Code	Refer to Toolkit References
14	DateOfBirth	Date of Birth	Format: DD/MM/YYYY
15	PlaceOfBirthArabic	Place of Birth in Arabic	

#	Attribute Name	Description	Remarks
16	PlaceOfBirthEnglish	Place of Birth in English	

2.2. Modifiable Data

2.2.1. Occupation Details

#	Attribute Name	Description	Remarks
1	OccupationCode	Occupation Code	Refer to Toolkit References
2	OccupationArabic	Occupation in Arabic	Refer to Toolkit References
3	OccupationEnglish	Occupation in English	Refer to Toolkit References
4	FamilyID	Family ID	
5	OccupationTypeArabic	Occupation Type in Arabic	Refer to Toolkit References
6	OccupationTypeEnglish	Occupation Type in English	Refer to Toolkit References
7	OccupationFieldCode	Occupation Field Code	Refer to Toolkit References

2.2.2. Company Details

#	Attribute Name	Description	Remarks
8	CompanyNameArabic	Company Name in Arabic	
9	CompanyNameEnglish	Company Name in English	

2.2.3. Marital Status

#	Attribute Name	Description	Remarks
10	MaritalStatusCode	Marital Status Code	Refer to Toolkit References
11	HusbandIdNumber	Husband Emirates ID Number	

2.2.4. Passport Details

#	Attribute Name	Description	Remarks
12	PassportNumber	Passport Number	
13	PassportTypeCode	Passport Type Code	Refer to Toolkit References
14	PassportExpiryDate	Passport Expiry Date	Format: DD/MM/YYYY
15	PassportIssueDate	Passport Issue Date	Format: DD/MM/YYYY

#	Attribute Name	Description	Remarks
16	PassportCountryCode	Passport Issuing Country Code	Refer to Toolkit References for Nationality codes
17	PassportCountryArabic	Passport Issuing Country in Arabic	
18	PassportCountryEnglish	Passport Issuing Country in English	

2.2.5. Sponsor Details

#	Attribute Name	Description	Remarks
19	SponsorType	Sponsor Type	Refer to Toolkit References
20	SponsorNumber	Sponsor Number	
21	SponsorName	Sponsor Name	
22	SponsorUnifiedNumber	Sponsor Unified Number	

2.2.6. Qualification Details

#	Attribute Name	Description	Remarks
23	ResidencyNumber	Residency Number	
24	ResidencyExpiryDate	Residency Expiry Date	Format: DD/MM/YYYY
25	QualificationLevelCode	Qualification Level Code	
26	QualificationLevelArabic	Qualification Level in Arabic	
27	QualificationLevelEnglish	Qualification Level in English	
28	DegreeDescriptionArabic	Degree Description in Arabic	
29	DegreeDescriptionEnglish	Degree Description in English	
30	FieldOfStudyCode	Field of Study Code	Refer to Toolkit References
31	FieldOfStudyArabic	Field of Study in Arabic	Refer to Toolkit References
32	FieldOfStudyEnglish	Field of Study in English	Refer to Toolkit References

2.3. Address Data

2.3.1. Home Address

#	Attribute Name	Description	Remarks
1	AddressTypeCode	Address Type Code	
2	LocationCode	Location Code	
3	EmiratesCode	Emirates Code	Refer to Toolkit References
4	EmiratesDescArabic	Emirate Description in Arabic	Refer to Toolkit References
5	EmiratesDescEnglish	Emirate Description in English	Refer to Toolkit References
6	CityCode	City Code	Refer to Toolkit References
7	CityDescArabic	City Description in Arabic	Refer to Toolkit References
8	CityDescEnglish	City Description in English	Refer to Toolkit References
9	StreetArabic	Street Description in Arabic	
10	StreetEnglish	Street Description in English	
11	AreaCode	Area Code	Refer to Toolkit References
12	AreaDescArabic	Area Description in Arabic	Refer to Toolkit References
13	AreaDescEnglish	Area Description in English	Refer to Toolkit References
14	BuildingNameArabic	Building Name in Arabic	
15	BuildingNameEnglish	Building Name in English	
16	POBox	P.O. Box Number	
17	FlatNo	Flat Number	
18	ResidentPhoneNumber	Resident Phone Number	
19	MobilePhoneNumber	Mobile Phone Number	
20	CompanyNameArabic	Company Name in Arabic	
21	CompanyNameEnglish	Company Name in English	

2.3.2. Work Address

Work Address has the same fields as Home Address listed above.

2.4. Photo

#	Attribute Name	Description	Remarks
1	Photo	Card Holder Photo Image	JPEG format

2.5. Signature Image

#	Attribute Name	Description	Remarks
1	HolderSignatureImage	Card Holder Hand Signature Image	TIFF format (Tagged Image File Format)

2.6. Family Book

Family Book contains information regarding the family head, up to four wives, and up to twenty children.

2.6.1. Head of Family Details

#	Attribute Name	Description	Remarks
1	HolderIdNumber	Family Head Emirates ID Number	
2	FamilyID	Family Identifier	
3	EmirateNameArabic	Emirate Name in Arabic	
4	EmirateNameEnglish	Emirate Name in English	
5	FirstNameArabic	First Name in Arabic	
6	FirstNameEnglish	First Name in English	
7	FatherNameArabic	Father Name in Arabic	
8	FatherNameEnglish	Father Name in English	
9	GrandFatherNameArabic	Grandfather Name in Arabic	
10	GrandFatherNameEnglish	Grandfather Name in English	
11	FamilyNameArabic	Family Name in Arabic	
12	FamilyNameEnglish	Family Name in English	

2.6.2. Wife Details (for each wife)

#	Attribute Name	Description	Remarks
1	WifeldNumber	Wife Emirates Identity Number	

#	Attribute Name	Description	Remarks
2	FullNameArabic	Full Name in Arabic	Same composition as Non-Modifiable FullNameArabic
3	FullNameEnglish	Full Name in English	Same composition as Non-Modifiable FullNameEnglish

2.6.3. Child Details (for each child)

#	Attribute Name	Description	Remarks
1	ChildIdNumber	Child Emirates Identity Number	
2	FullNameArabic	Full Name in Arabic	
3	FullNameEnglish	Full Name in English	
4	Gender	Gender	M – Male, F – Female
5	DateOfBirth	Date of Birth	Format: DD/MM/YYYY
6	PlaceOfBirthArabic	Place of Birth in Arabic	
7	PlaceOfBirthEnglish	Place of Birth in English	
8	MotherIdNumber	Mother Emirates Identity Number	
9	MotherFullNameArabic	Mother Full Name in Arabic	
10	MotherFullNameEnglish	Mother Full Name in English	

3. Procedure for Certificate Preparation

This section describes the preparation procedure for the X.509 certificates used by the TLS protocol.

A certificate chain is a combination of all related certificates together in a single file commonly called certificate bundle or chain. Certificate verification happens against the certificate chain or bundles. For preparing the certificate chain, the related certificates need to be in Base64-encoded X.509 format.

3.1. Preparing Base64 X.509 Certificate

Base64 encoded X.509 certificates are in text format with the following structure:

```
-----BEGIN CERTIFICATE-----  
<Base64 Encoded Certificate Data>  
-----END CERTIFICATE-----
```

If the certificates are not in PEM format, the following OpenSSL command can be used to convert from DER to PEM format:

```
openssl x509 -inform der -in certificate.cer -out certificate.pem
```

3.2. Preparing Certificate Chain

The certificate chain is prepared by concatenating related Base64 X.509 certificates in a single file:

```
cat root-ca.pem intermediate-ca.pem > certificate-chain.pem
```

4. Configuration Parameters

There are two approaches to configure the Toolkit SDK:

1. **Programmatic Configuration** – this approach supports specifying configuration parameters directly through the toolkit SDK API. The name of the parameter through which the configuration is provided is `config_params`.
2. **Declarative Configuration** – this approach supports specifying configuration parameters through an XML-based configuration file known as `config_ap`. The ICP provides the `config_ap` file along with all other required configuration files.

The following table describes the behavior of the Toolkit during initialization based on the two configuration sources:

<code>config_params</code>	<code>config_ap</code>	Behavior
Provided	Provided	Parameters from <code>config_params</code> and <code>config_ap</code> are consolidated. Where a parameter is provided by both sources, the value from <code>config_params</code> takes precedence.
Provided	Not Provided	Parameters in <code>config_params</code> are used. The toolkit searches for all required configuration files in the directory where the toolkit binaries are located. If not found, the Toolkit will not initialize.
Not Provided	Provided	Parameters in <code>config_ap</code> are used.
Not Provided	Not Provided	The toolkit searches for all required configuration files in the directory where the toolkit binaries are located. If all required files are found, the toolkit will complete initialization. Once initialized, only the offline validation mode is available.

4.1. Configuration Parameters (`config_ap` / `config_params`)

Parameter	Description
<code>config_ag</code>	Path to Toolkit Agent configuration file (<code>config_ag</code>). This is the primary Toolkit configuration file and is a mandatory parameter when operating in online mode.
<code>config_pd</code>	Path to a Public Data configuration file (<code>config_pd</code>). This file determines the scope of data to be read from the card. If not provided, a default configuration is used.
<code>config_url</code>	URL path from where configuration files are downloaded.
<code>config_directory</code>	Directory path from where configuration files are loaded. If <code>config_url</code> is specified along with this parameter, then this parameter will be ignored.

Parameter	Description
environment	Used to indicate a sub-path under config_url and config_directory where all required configuration files (except config_ap) are loaded from.
log_directory	Overrides the default log directory to store toolkit debug log files.
log_level	Controls the level of detail in the log file. Valid values: 0 (Off), 1 (Error), 2 (Debug), 3 (Verbose).
enable_digital_signature_service	When set to "true", the Toolkit initializes the DSS engine required for digital signature operations (PAdES, XAdES, CAdES).
jre32_directory / jre64_directory	Path to the 32/64-bit Java runtime environment (JRE). Required when enable_digital_signature_service is set to true.
agent_tls_enabled	Enables TLS mode for the Toolkit Agent service. Valid values: "true" or "false".
agent_ip_address	Specifies the IP address of the Toolkit Agent service. Default is "127.0.0.1".
agent_port	Specifies the port number of the Toolkit Agent service.
vg_connection_timeout	Sets the VG request timeout in seconds. If also provided in config_ag, this value takes precedence.

4.2. Configuration Parameters (config_ag_ap)

Parameter	Description
agent_init_retry_interval	Toolkit Agent Service only: periodic wait time in seconds for initialization retry on network error.
application_session_timeout	Toolkit Agent Service only: timeout period in minutes for an application session.
cache_timeout	Toolkit Agent Service only: maximum period in minutes for cache entries to remain valid.

5. Request ID

The purpose of the Request ID is to bind a Toolkit response to its corresponding request. This is an effective security measure to prevent the replay of old responses.

The Request ID is a randomly generated sequence of 40 bytes. An application that integrates with the ID Card Toolkit is responsible for generating the Request ID. A new request ID must be generated for each request. It must be ensured that a reliable secure random number generator is used in the generation process. The Request ID should be encoded in Base64 format and provided to the Toolkit service calls.

After receiving a Toolkit service response, the application must verify the response signature and confirm that the Request ID in the response matches the one that was sent.

6. PIN Encoding Procedure

The PIN must be encrypted before it is sent to the Toolkit. The procedure is as follows:

1. Construct a byte array by concatenating:
 - First 8 bytes: the Request Handle received from a successful call to `PrepareRequest`, Base64-decoded.
 - Remaining bytes: PIN digits (0–9) appended after the Request Handle.
 - Total PIN digits cannot exceed 16; minimum is 4.
2. Encrypt the byte array using RSA encryption with the VG public key (2048-bits) and PKCS#1 padding. The VG public key is obtained from a successful call to `GetDataProtectionKey`.
3. The encrypted output is 256 bytes.
4. Convert the 256-byte encrypted result to Base64 encoding.
5. The Base64-encoded value is the encrypted PIN to be provided to the Toolkit.

7. User ID & Password Encoding Procedure

The User ID/Password must be encrypted before it is sent to the Toolkit. The procedure is the same as the PIN encoding, except:

1. Construct a byte array by concatenating:
 - First 8 bytes: the Request Handle (Base64-decoded).
 - Remaining bytes: User ID/Password appended after the Request Handle.
2. Encrypt using RSA with VG public key (2048-bits) and PKCS#1 padding.
3. Convert the 256-byte encrypted result to Base64 encoding.

8. Fingerprint Reference Identifiers

The following table lists the fingerprint reference identifiers used for biometric authentication:

Identifier	Description
RIGHT_THUMB	Right Thumb
RIGHT_INDEX	Right Index Finger
RIGHT_MIDDLE	Right Middle Finger
RIGHT_RING	Right Ring Finger
RIGHT_LITTLE	Right Little Finger
LEFT_THUMB	Left Thumb
LEFT_INDEX	Left Index Finger
LEFT_MIDDLE	Left Middle Finger
LEFT_RING	Left Ring Finger
LEFT_LITTLE	Left Little Finger
UNKNOWN	Unknown Finger (Toolkit determines the finger)

9. Digital Signature Context Attributes

The following parameters are used for configuring digital signature operations:

Parameter	Description
SIGNATURE_FORMAT	Type of digital signature: XADES – XML Advanced Electronic Signatures PADES – PDF Advanced Electronic Signatures CADES – CMS Advanced Electronic Signatures
SIGNATURE_LEVEL	Level of digital signature: BASELINE_B – Basic signature BASELINE_T – Adding trusted timestamp BASELINE_LT – Adding long term validation data BASELINE_LTA – Adding archived timestamp for long term archival
PACKAGING_MODE	Packaging mode (per ETSI EN 319 162): ENVELOPED – Signature surrounds the data ENVELOPING – Signed data is within the signature DETACHED – Signature is separate from the data
DIGEST_ALGORITHM	Hash algorithm: SHA256 – SHA-256 SHA512 – SHA-512
TSP_URL	Timestamp Authority URL
CRL_URL	Certificate Revocation List URL
OCSP_URL	Online Certificate Status Protocol URL

10. PAdES Signature Parameters

Additional parameters for PDF visible signatures:

Parameter	Description
name	Signer's name to display on the visible signature
name_x	X-coordinate position of the signature block on the page
name_y	Y-coordinate position of the signature block on the page
name_w	Width of the signature block
name_h	Height of the signature block
name_position	Position of the signature text relative to the signature (LEFT, RIGHT, TOP, BOTTOM)
page_number	Page number of the PDF on which to place the visible signature

11. Digital Signature Verification Context Attributes

Parameter	Description
ocsp_url	OCSP URL for certificate status checking
cert_path	Folder path containing issuer and root certificates
is_detached	TRUE if the verified document is of detached type, otherwise FALSE

12. Digital Signature Verification Report Types

The Toolkit supports three types of verification reports as defined by the ETSI standard.

12.1. Simple Report

The Simple Report provides a summary of the validation result. It contains the main information about the signatures and their validity. The result is presented in a simplified format that allows end users to quickly understand the validation outcome.

The simple report includes:

- Signature validity status (TOTAL_PASSED, TOTAL_FAILED, INDETERMINATE)
- Signer information
- Signature format and level
- Certificate chain validity

12.2. Detailed Report

The Detailed Report provides a comprehensive validation result based on the ETSI standard. It is built around Basic Building Blocks (BBB) which include:

- **ISC** – Identification of the Signing Certificate
- **VCI** – Validation Context Initialization
- **CV** – Cryptographic Verification
- **SAV** – Signature Acceptance Validation
- **XCV** – X.509 Certificate Validation

Each block contains rules executed sequentially. If all rules pass, the conclusion indicates PASSED. Otherwise FAILED or INDETERMINATE is returned.

12.3. Diagnostic Data

The Diagnostic Data report provides the most detailed information about all certificates, timestamps, and signatures involved in the validation process. It includes:

- Certificate details (subject, issuer, validity, key usage)
- Timestamp information
- Revocation data (CRL, OCSP)
- Signature scopes and cryptographic parameters

13. Error Codes

The following tables list all error codes returned by the Toolkit (367 codes). Error codes are defined in `EIDAToolkitError.h`. Use `GetStatusMessage(status)` to get a human-readable description at runtime.

13.1. General Errors

Code	Name	Description
0	ETSTATUS_SUCCESS	The operation completed successfully

13.2. General Errors (1-29)

Code	Name	Description
1	ETSTATUS_PLUGIN_ERROR	Plugin operation failed
2	ETSTATUS_INVALID_PARAMETER	Invalid parameter
3	ETSTATUS_NOT_INITIALIZED	The module or object is not initialized
4	ETSTATUS_ALREADY_INITIALIZED	The module or object is already initialized
5	ETSTATUS_INVALID_HANDLE	Invalid handle value
6	ETSTATUS_INVALID_DATA_TYPE	Invalid data type
7	ETSTATUS_INVALID_DATA	Invalid data
8	ETSTATUS_INSUFFICIENT_DATA	Insufficient data
9	ETSTATUS_INVALID_COMMAND	Invalid command code used
10	ETSTATUS_MALLOC_FAILED	Memory allocation failed
11	ETSTATUS_FILE_NOT_FOUND	Requested file not found
12	ETSTATUS_COULD_NOT_OPEN_FILE	File could not be opened
13	ETSTATUS_FILE_EMPTY	Specified file is empty
14	ETSTATUS_FILE_SEEK_FAILED	File seek operation failed
15	ETSTATUS_FILE_READ_FAILED	File read operation failed
16	ETSTATUS_FILE_WRITE_FAILED	File write operation failed
17	ETSTATUS_INDEX_OUT_OF_BOUNDS	Index is out of bounds
18	ETSTATUS_NOT_FOUND	Searched object not found

Code	Name	Description
19	ETSTATUS_MORE_DATA_AVAILABLE	More data remaining to be read
20	ETSTATUS_INSUFFICIENT_BUFFER_LEN	Buffer provided is of insufficient length
21	ETSTATUS_BAD_FORMAT	Provided data is of invalid format
22	ETSTATUS_ERROR_NETWORK	Network error occurred
23	ETSTATUS_NOT_SUPPORTED	Functionality not supported
24	ETSTATUS_BLUETOOTH_NOT_SUPPORTED	Bluetooth not supported
25	ETSTATUS_BLUETOOTH_IS_OFF	Bluetooth is off. Please check the settings
26	ETSTATUS_NO_PAIRING_DEVICES	Matching paired device not found
27	ETSTATUS_SERVICE_NOT_AUTHORIZED	Toolkit service usage not authorized as per the license
28	ETSTATUS_LICENSE_DATA_CORRUPTED	Toolkit license file is corrupt
29	ETSTATUS_OBJECT_NOT_INITIALIZED	Uninitialized object
30	ETSTATUS_NULL_OBJECT_POINTER	Object reference is null
31	ETSTATUS_OBJECT_ATTRIBUTE_NOT_FOUND	Requested attribute not found in the object
32	ETSTATUS_OBJECT_ATTRIBUTE_CORRUPTED	Requested object attribute found corrupted
33	ETSTATUS_NULL_OBJECT_ATTRIBUTE	Object attribute reference is null

13.3. Agent & Schema Errors (34-38)

Code	Name	Description
34	ETSTATUS_NO_MORE_FREE_CONTEXTS	Agent has reached the maximum limit for application connections
35	ETSTATUS_NO_APPLICATIONS_MAPPED	The schema contains no application maps
36	ETSTATUS_RAW_DATA_SCHEMA_MISMATCH	Raw data is not matching the ID Card schema
37	ETSTATUS_SYMBOL_NOT_FOUND	Failed to get function address from a shared library
38	ETSTATUS_ERROR_LOAD_MODULE	Failed to load a shared library

13.4. Device, Plugin & Smartcard Errors (39-69)

Code	Name	Description
39	ETSTATUS_SMARTCARD_ERROR	Smartcard operation failed
40	ETSTATUS_INVALID_SCARD_READER_CONFIG	Invalid or incomplete smartcard reader plugin configuration
41	ETSTATUS_INVALID_FPSCANNER_CONFIG	Invalid or incomplete fingerprint scanner plugin configuration
42	ETSTATUS_INVALID_SCARD_READER_PLUGIN	Smartcard reader plugin don't have the required functions
43	ETSTATUS_INVALID_FPSCANNER_PLUGIN	Fingerprint scanner plugin don't have the required functions
44	ETSTATUS_NO_SMARTCARD_PLUGINS	No smartcard reader plugins found
45	ETSTATUS_NO_FINGERPRINT_PLUGINS	No fingerprint scanner plugins found
46	ETSTATUS_INVALID_SCARD_READER_NAME	Invalid smartcard reader name
47	ETSTATUS_PLUGIN_CONTEXT_CORRUPTED	Device plugin context is corrupted
48	ETSTATUS_FPSCANNER_ALREADY_CONNECTED	Already connected to fingerprint scanner
49	ETSTATUS_FP_OPENDEVICE_ERROR	Failed to open fingerprint sensor
50	ETSTATUS_CAPTURE_IMAGE_ERROR	Failed to capture fingerprint image
51	ETSTATUS_DEVICE_ENUM_FAILED	Failed to enumerate connected devices
52	ETSTATUS_LIST_READERS_ERROR	Failed to list smartcard readers
53	ETSTATUS_SC_CONNECT_FAILED	Failed to establish connection with smartcard
54	ETSTATUS_SC_DISCONNECT_FAILED	Failed to close an established smartcard connection
55	ETSTATUS_SMARTCARD_NOT_PRESENT	Smartcard not present in the reader
56	ETSTATUS_PAIRIED_DEVICES_NOT_IN_RANGE	Paired device not in range
57	ETSTATUS_DEVICE_ERROR	Device error

Code	Name	Description
58	ETSTATUS_NO_SUPPORTED_DEVICE	No supported device found
59	ETSTATUS_CONNECTION_TIMEOUT	Device connection timed out
60	ETSTATUS_ERROR_VERIFY_ISO_TEMPLATE	Failed to verify fingerprint ISO template
61	ETSTATUS_ERROR_VERIFY_MATCH	Error occurred in matching fingerprint ISO templates
62	ETSTATUS_ESTABLISH_RM_CTXT_ERROR	Failed to establish resource manager context
63	ETSTATUS_RM_CTXT_CLOSE_FAILED	Failed to close established resource manager context
64	ETSTATUS_SEND_SVC_REQ_SC_ERROR	Failed to send smartcard command
65	ETSTATUS_SC_ATR_ERROR	Error in retrieving ATR from smartcard
66	ETSTATUS_INVALID_NFC_INTERFACE	Not an NFC interface
67	ETSTATUS_CONNECTION_LIMIT_REACHED	Smartcard maximum connection limit reached
68	ETSTATUS_CREATE_WS_CONTEXT_FAILED	Failed to create websocket context
69	ETSTATUS_WS_SVC_FAILED	Failed to run websocket service

13.5. Network & VG Communication Errors (70-114)

Code	Name	Description
70	ETSTATUS_ERROR_QUERY_HEADER	Error in preparing HTTP request header
71	ETSTATUS_ERROR_VGURL_NOT_PROVIDED	Validation Gateway (VG) URL is not provided
72	ETSTATUS_ERROR_VGURL_NOT_SECURE	Validation Gateway (VG) URL provided is not secure
73	ETSTATUS_VG_DISCONNECTED	Validation Gateway (VG) is disconnected
74	ETSTATUS_CARD_REVOKED	Emirates ID Card is revoked
75	ETSTATUS_CARD_EXPIRED	Emirates ID Card is expired

Code	Name	Description
76	ETSTATUS_BAD_VG_REQUEST	Bad Validation Gateway (VG) request
77	ETSTATUS_BIO_AUTH_FAILED	Low score match
78	ETSTATUS_DATASIGNATURE_ERROR	Failed to validate public data signature
79	ETSTATUS_PINVERIFICATION_FAILED	Emirates ID Card PIN verification failed
80	ETSTATUS_PARSE_FP_ERROR	Failed to parse fingerprint data from card
81	ETSTATUS_ERROR_GET_TEMPLATES_COUNT	Failed to retrieve fingerprint templates count
82	ETSTATUS_ERROR_RETRIEVING_TEMPLATE	Failed to retrieve fingerprint templates
83	ETSTATUS_NO_BIO_DATA	There is no biometric data present in the Emirates ID Card
84	ETSTATUS_CARD_BLOCKED	Emirates ID Card is blocked
85	ETSTATUS_VERIFYSIGNATURE_ERROR	Failed to verify digital signature
86	ETSTATUS_UNKNOWN_SMARTCARD	Unknown smartcard
87	ETSTATUS_INVALID_CHECK_BIT	Error in computing check bit
88	ETSTATUS_WRONG_MAC	Invalid Message Authentication Code (MAC) data
89	ETSTATUS_MOC_SEC_STATUS_UNSATISFIED	Match on Card (MOC) security status not satisfied
90	ETSTATUS_MOC_APPLICATION_BLOCKED	Match on Card (MOC) applet is blocked
91	ETSTATUS_MOC_MATCH_BAD_TEMPLATE	Invalid fingerprint template data
92	ETSTATUS_MOC_FAILED	Match on Card (MOC) failed
93	ETSTATUS_MOC_MATCH_STATUS_UNKNOWN	Match on Card (MOC) status unknown
94	ETSTATUS_CARD_MUTUAL_AUTH_FAILED	BAC authentication parameters are not correct
95	ETSTATUS_PKI_APPLICATION_BLOCKED	PKI application is blocked
96	ETSTATUS_PKI_PIN_BLOCKED	PKI PIN is blocked

Code	Name	Description
97	ETSTATUS_PKI_PIN_VERIFY_SEC_UNKNOWN	PKI PIN verify security status not satisfied
98	ETSTATUS_PKI_PIN_BAD_REFERENCE	Invalid PKI PIN reference
99	ETSTATUS_PKI_PIN_VERIFY_STATUS_UNKNOWN	Invalid PKI PIN verification status
100	ETSTATUS_VGREQUEST_IS_NOT_VALID	Request is not according to Validation Gateway (VG) Schema
101	ETSTATUS_VG_LICENSE_INVALID	The license key is invalid
102	ETSTATUS_VG_GEN_PKI_SESSIONKEY_FAILED	Failed to generate PKI session keys
103	ETSTATUS_COMPLETE_AUTH_FAILED	Failed to complete PKI authentication
104	ETSTATUS_CARD_IS_NOT_GENUINE	The Emirates ID Card is not genuine
105	ETSTATUS_CIPHERED_PIN_VERIFY_FAILED	Ciphered PIN verification failed
106	ETSTATUS_APPLICATION_BLOCKED	Application is blocked
107	ETSTATUS_SERVICE_LIC_INVALID	The license does not grant access to the requested service
110	ETSTATUS_PIN_RESET_FAILED	Failed to reset Emirates ID Card PKI PIN
111	ETSTATUS_VG_RESPONSE_EXPIRED	Received expired Validation Gateway (VG) response
112	ETSTATUS_COMPLETE_TRXN_FAILED	Failed to complete transaction successfully
113	ETSTATUS_NO_FAMILY_BOOK_DATA	No family book data present in the Emirates ID Card
114	ETSTATUS_INVALID_VG_RESPONSE	Invalid Validation Gateway (VG) response

13.6. Digital Signature Errors (115-120)

Code	Name	Description
115	ETSTATUS_XADES_SIGNING_FAILED	Failed to sign document using XAdES standard

Code	Name	Description
116	ETSTATUS_PADES_SIGNING_FAILED	Failed to sign document using PAdES standard
117	ETSTATUS_CADES_SIGNING_FAILED	Failed to sign document using CAdES standard
118	ETSTATUS_XADES_VERIFY_FAILED	Failed to verify document using XAdES standard
119	ETSTATUS_PADES_VERIFY_FAILED	Failed to verify document using PAdES standard
120	ETSTATUS_CADES_VERIFY_FAILED	Failed to verify document using CAdES standard

13.7. Certificate & Crypto Errors (121-135)

Code	Name	Description
121	ETSTATUS_CERT_VERIFY_CONTEXT_NULL	Failed to create certificate verification context
122	ETSTATUS_CERT_VERIFY_STATUS_UNSPECIFIED	Unspecified certificate validation response
123	ETSTATUS_CERT_NOT_YET_VALID	Certificate not valid yet
124	ETSTATUS_CERT_HAS_EXPIRED	Certificate expired
125	ETSTATUS_TLS_CERT_INVALID	Server TLS certificate validation failed
126	ETSTATUS_CERT_REVOKED	The certificate is revoked
127	ETSTATUS_CERT_ISSUER_NOT_FOUND	Certificate issuer not found in the local store
128	ETSTATUS_SSL_CERT_NOT_RETRIEVED	Error in retrieving SSL certificate
129	ETSTATUS_GEN_SYMMETRIC_KEY_ERROR	Failed to generate symmetric key
130	ETSTATUS_ERROR_PARSING_X509_CERT	Failed to parse x509 certificate
131	ETSTATUS_ERROR_GET_CERT_SLNO	Failed to get certificate serial number
132	ETSTATUS_GET_PUBKEY_FAILED	Failed to retrieve public key from certificate
133	ETSTATUS_ERROR_GENERATING_DIGEST	Failed to generate message digest

Code	Name	Description
134	ETSTATUS_GET_RSA_PUBKEY_FAILED	Failed to get RSA public key
135	ETSTATUS_ERROR_APPEND_BIO	Failed to append SSL BIO

13.8. Java/JVM Errors (136, 200-207, 293-294)

Code	Name	Description
136	ETSTATUS_JAVA_VM_CREATION_FAILED	Failed to load Java Virtual Machine (VM)
137	ETSTATUS_PIN_NOT_CREATED	PIN is not created

13.9. Configuration & License Errors (138-139, 187-189, 224-244, 296, 363-366)

Code	Name	Description
138	ETSTATUS_UNSIGNED_CONFIG_IN_PROD	Unsigned config file used in production
139	ETSTATUS_INVALID_SP	Service Provider (SP) is not authorized

13.10. Card & PKI Authentication Errors (140-144)

Code	Name	Description
140	ETSTATUS_CARD_UNKNOWN	The Emirates ID Card is unknown
141	ETSTATUS_PKI_AUTH_FAILED	PKI authentication failed
142	ETSTATUS_UNBLOCK_PIN_FAILED	Failed to unblock PIN
143	ETSTATUS_DSS_ERROR	Digital Signature Service (DSS) internal error
144	ETSTATUS_INVALID_PROTOCOL	Invalid protocol

13.11. VG Server-Side Errors (145-186)

Code	Name	Description
145	ETSTATUS_VG_SYSTEM_ERROR	Validation Gateway (VG) system error (E500-00-00)
146	ETSTATUS_SESSION_NOT_INITIALIZED	Session not initialized (E400-01-01)
147	ETSTATUS_MISSING_PARAMETER	Required parameter missing (E400-01-02)
148	ETSTATUS_VG_INVALID_PARAMETER	Invalid parameter value (E400-01-03)

Code	Name	Description
149	ETSTATUS_REPOSITORY_SYSTEM_ERROR	Repository system error (E500-02-00)
150	ETSTATUS_WIRE_FMT_CHECKSUM	Invalid wire format checksum (E400-40-01)
151	ETSTATUS_INVALID_WIRE_FMT_FIRST_BYTE	Invalid wire format first byte (E400-40-02)
152	ETSTATUS_INVALID_WIRE_FMT_HEADER_LEN	Invalid wire format header length (E400-40-03)
153	ETSTATUS_INVALID_WIRE_FMT_TOTAL_LEN	Invalid wire format total length (E400-40-04)
154	ETSTATUS_INVALID_PAYLOAD_ENCRYPTION	Invalid payload encryption (E400-40-06)
155	ETSTATUS_INVALID_PAYLOAD_XML_MSG	Invalid payload XML message (E400-42-02)
156	ETSTATUS_INVALID_PAYLOAD_XML_SIGNATURE	Invalid payload XML signature (E400-42-03)
157	ETSTATUS_PAYLOAD_UNMARSHAL_ERROR	Payload unmarshal error (E400-42-04)
158	ETSTATUS_PAYLOAD_SIGNATURE_NOT_VERIFIED	Payload signature not verified (E400-42-05)
159	ETSTATUS_INVALID_SERVICE_PROVIDER	Service provider not found (E400-44-01)
160	ETSTATUS_SERVICE_MISMATCH	Service mismatch (E400-44-02)
161	ETSTATUS_CSN_MISMATCH	Card serial number mismatch (E400-44-03)
162	ETSTATUS_CARD_NOT_REGISTERED	The Emirates ID Card is not registered (E400-44-04)
163	ETSTATUS_ID_NUMBER_MISMATCH	ID number mismatch (E400-44-05)
164	ETSTATUS_UNSUPPORTED_SDK_VERSION	Unsupported SDK version (E400-44-06)
165	ETSTATUS_FINGERPRINT_INTEGRITY_ERROR	Fingerprint integrity error (E400-54-01)
166	ETSTATUS_DEVICE_ALREADY_REGISTERED	Device already registered (E409-73-01)

Code	Name	Description
167	ETSTATUS_LIC_KEY_INVALID	License key invalid (E401-72-01)
168	ETSTATUS_LIC_PERIOD_NOT_STARTED	License validity period not started (E401-72-02)
169	ETSTATUS_LIC_PERIOD_ENDED	License validity period ended (E401-72-03)
170	ETSTATUS_SERVICE_NOT_LICENSED	Service not licensed (E401-72-04)
171	ETSTATUS_DEVICE_NOT_REGISTERED	Device not registered (E404-73-01)
172	ETSTATUS_INVALID_DATA_LEN	Invalid data length (E400-70-01)
173	ETSTATUS_INVALID_DATA_NONCE	Invalid data nonce (E400-70-02)
174	ETSTATUS_INVALID_FP_TEMPLATE	Invalid fingerprint template (E400-71-01)
175	ETSTATUS_INVALID_PAYLOAD_XML	Invalid payload XML (E400-42-01)
176	ETSTATUS_VG_INTEGRITY_ERROR	Integrity of the request could not be validated (E400-63-01)
177	ETSTATUS_INVALID_ENCRYPTION_DATA	Incorrect encryption data (E400-70-03)
178	ETSTATUS_NULL_PIN	Invalid PIN, values must not be null (E400-74-01)
179	ETSTATUS_INVALID_PIN_LENGTH	Invalid PIN, length must be between 4 and 16 digits [0-9] (E400-74-02)
180	ETSTATUS_INVALID_PIN_DATA	Invalid PIN, values must only contain digits between 0 and 9 (E400-74-03)
181	ETSTATUS_INVALID	Service provider not found (E400-44-11)
182	ETSTATUS_INVALID_SERVICE	Service mismatch (E400-44-12)
183	ETSTATUS_CSN_INVALID	Card serial number mismatch (E400-44-13)

Code	Name	Description
184	ETSTATUS_CN_INVALID	Card number mismatch (E400-44-14)
185	ETSTATUS_IDN_INVALID	ID number mismatch (E400-44-15)
186	ETSTATUS_SDK_VER_INVALID	Unsupported SDK version (E400-44-16)
187	ETSTATUS_INVALID_SERVICE_CTXT	Invalid service context
188	ETSTATUS_INVALID_LIC	Invalid config (config_li) data
189	ETSTATUS_INVALID_CONFIG_DATA	Invalid or incomplete configuration data
190	ETSTATUS_AUTH_PARAMS_NOT_SET	Authentication parameters not set
191	ETSTATUS_EXECUTE_SM_CMD_FAILED	Failed to execute Secure Messaging (SM) command
192	ETSTATUS_ERROR	The operation failed
193	ETSTATUS_NO_SUPPORTED_READER	No supported reader detected
194	ETSTATUS_NFC_TAG_INVALID	Invalid NFC tag
195	ETSTATUS_NFC_CONN_ERROR	NFC connection error
196	ETSTATUS_INVALID_PLUGIN_CONTEXT	Invalid plugin context
197	ETSTATUS_NO_CONN_USB_DEV	No USB device connected
198	ETSTATUS_ERROR_TIMEOUT	Network timeout occurred

13.12. Plugin & Runtime Errors (199-210)

Code	Name	Description
199	ETSTATUS_NO_SMARTCARD_PLUGINS_LOADED	No smartcard plugins loaded
200	ETSTATUS_LOAD_JAVA_CLASS_FAILED	Failed to load Java class
201	ETSTATUS_GET_METHODID_FAILED	Failed to get method ID of Java class
202	ETSTATUS_NEW_STRING_ERROR	Failed to construct new string object
203	ETSTATUS_NEW_OBJECT_ERROR	Failed to construct new object

Code	Name	Description
204	ETSTATUS_GET_FIELD_ID_FAILED	Failed to retrieve field ID from Java object
205	ETSTATUS_GET_INT_FIELD_FAILED	Failed to retrieve int field from Java object
206	ETSTATUS_GET_OBJECT_FIELD_FAILED	Failed to retrieve object field from Java object
207	ETSTATUS_FAILED_CALLING_OBJ_METHOD	Failed to call method from Java object
208	ETSTATUS_DECRYPT_SIGNATURE_FAILED	Failed to decrypt signature
209	ETSTATUS_SET_CONTEXT_FAILED	Failed to set Android application context
210	ETSTATUS_PLUGIN_INIT_FAILED	Plugin initialization failed

13.13. OCSP & Certificate Status Errors (211-221)

Code	Name	Description
211	ETSTATUS_OCSP_RESP_MALFORMEDREQUEST	Malformed OCSP request
212	ETSTATUS_OCSP_RESP_INTERNALERROR	OCSP internal error
213	ETSTATUS_OCSP_RESP_TRYLATER	OCSP response - try later
214	ETSTATUS_OCSP_RESP_SIGREQUIRED	OCSP response- SIGREQUIRED
215	ETSTATUS_OCSP_RESP_UNAUTHORIZED	OCSP response - unauthorized
216	ETSTATUS_CERT_STATUS_REVOKED	Certificate status - revoked
217	ETSTATUS_CERT_STATUS_UNKNOWN	Certificate status - unknown
218	ETSTATUS_INVALID_CERT_TYPE	Invalid certificate type
219	ETSTATUS_CAPTURE_ISO_ERROR	Failed to capture fingerprint template
220	ETSTATUS_INVALID_PLATFORM	Platform not supported
221	ETSTATUS_INVALID_JVM_PATH	Invalid JVM path
222	ETSTATUS_APPLET_NOT_PERMITTED	Unauthorized use of applet mode

Code	Name	Description
223	ETSTATUS_NO_SERVER_TLS_CERT	License file contains invalid/ no server TLS certificate & chain
224	ETSTATUS_INVALID_VG_QA	Invalid config (config_vg_qa) data
225	ETSTATUS_INVALID_VG_PROD	Invalid config (config_vg_prod) data
226	ETSTATUS_INVALID_LV_QA	Invalid config (config_lv_qa) data
227	ETSTATUS_INVALID_LV_PROD	Invalid config (config_lv_prod) data
228	ETSTATUS_INVALID_TK_QA	Invalid config (config_tk_qa) data
229	ETSTATUS_INVALID_TK_PROD	Invalid config (config_tk_prod) data
230	ETSTATUS_INVALID_PG	Invalid config (config_pg) data
231	ETSTATUS_INVALID_AG	Invalid config (config_ag) data
232	ETSTATUS_LICENSE_CONTAIN_NO_SERVICE	License contains no service
233	ETSTATUS_SERVER_RESPONSE_ERROR	Failed to get response from server
234	ETSTATUS_VG_CONNECTION_ERROR	Validation Gateway (VG) connection error
235	ETSTATUS_NETWORK_CONN_CLOSED	Server has closed connection unexpectedly
236	ETSTATUS_INVALID_CONFIG_PEER_CERT	Config TLS certificate's peer certificate is not valid
237	ETSTATUS_ERROR_CONFIG_URL_NOT_SECURE	Config URL provided is not secure
238	ETSTATUS_ERROR_CONFIG_CERT_NOT_PROVIDED	Config TLS certificate is not provided
239	ETSTATUS_ERROR_DEV_ID	Failed to retrieve device ID
241	ETSTATUS_NO_CONFIG_URL	Config URL is not provided
242	ETSTATUS_CFG_LI_VALIDATION_ERROR	Failed to validate config_li signature

Code	Name	Description
243	ETSTATUS_CFG_VG_VALIDATION_ERROR	Failed to validate config_vg signature
244	ETSTATUS_CFG_PD_VALIDATION_ERROR	Failed to validate config_pd signature

13.14. Smartcard Reader Errors (245-261)

Code	Name	Description
245	ETSTATUS_SC_REMOVED	Smartcard has been removed, please insert the card & connect again
246	ETSTATUS_SC_OP_CANCELED	Smartcard operation was canceled
247	ETSTATUS_SC_RESET	Smartcard was reset
248	ETSTATUS_SC_ACCESS_DENIED	Smartcard access was denied because of a security violation
249	ETSTATUS_SC_UNPOWERED	Power has been removed from smartcard
250	ETSTATUS_SC_NOT_RESPONDING	Smartcard is not responding to reset
251	ETSTATUS_SC_ATR_CONFLICT	Cannot communicate with smartcard due to ATR conflict
252	ETSTATUS_SC_CONN_ERROR	Smartcard cannot be accessed due to other outstanding connections
253	ETSTATUS_SC_COMM_ERROR	Detected communication error with smartcard
254	ETSTATUS_SC_READER_DRIVER_ERR	Smartcard reader driver error
255	ETSTATUS_NO_SC_READERS	No smartcard readers available
256	ETSTATUS_SC_RES_MANAGER_STOPPED	Smartcard resource manager is not running
257	ETSTATUS_NO_SC_IN_READER	No smartcard present in the reader
258	ETSTATUS_SC_READER_NOT_READY	Smartcard or reader is not ready
259	ETSTATUS_READER_NOT_AVAILABLE	Reader is not available for use
260	ETSTATUS_READER_NOT_RECOGNISED	Specified reader name is not recognized
261	ETSTATUS_UNRESPONSIVE_CARD	Smartcard is not responding to reset

13.15. JSON & XML Parsing Errors (262-282)

Code	Name	Description
262	ETSTATUS_JSON_ARRAY_EMPTY	Received empty JSON array
263	ETSTATUS_JSON_ARRAY_GET_ERROR	Failed to retrieve element from JSON array
264	ETSTATUS_JSON_NON_OBJECT_TYPE	Received JSON non object type where object is expected
265	ETSTATUS_JSON_CREATE_FAILED	Could not create new JSON object
266	ETSTATUS_JSON_ADD_FAILED	Could not add attributes to JSON object
267	ETSTATUS_UNSUPPORTED_JSON_DATA_TYPE	Provided JSON object type is not supported
268	ETSTATUS_UNEXPECTED_JSON_TYPE	JSON attribute data type is different from expected
269	ETSTATUS_XML_PARSE_ERROR	Error occurred while parsing XML
270	ETSTATUS_XML_GET_ROOT_ERROR	Error occurred while accessing the XML root node
271	ETSTATUS_XML_NODE_NOT_FOUND	Could not find the searched XML node
272	ETSTATUS_XML_NODE_EMPTY	XML node does not contain any values
273	ETSTATUS_XML_ATTRIBUTE_NOT_FOUND	XML node does not contain the searched attribute
274	ETSTATUS_XMLSEC_INIT_FAILED	Failed to initialize xmlsec library
275	ETSTATUS_XMLSEC_VER_CHECK_FAILED	Failed to check xmlsec library version
276	ETSTATUS_INIT_CRYPTOP_APP_ERROR	Failed to initialize xmlsec crypto app
277	ETSTATUS_INIT_CRYPTOP_LIB_ERROR	Failed to initialize xmlsec crypto library
278	ETSTATUS_XMLSEC_GET_NODE_ERROR	Failed to find xmlsec signature node
279	ETSTATUS_XMLSEC_SIG_CTXT_ERROR	Failed to generate signature context
280	ETSTATUS_LOAD_XMLSEC_PUB_KEY_FAILED	Failed to load xmlsec public key

Code	Name	Description
281	ETSTATUS_XMLSEC_DSIG_CTXT_ERROR	Xmlsec digital signature context error
282	ETSTATUS_XMLSEC_SIG_VERIFY_FAILED	Digital signature verification failed
283	ETSTATUS_OBJECT_ALREADY_ADDED	Object already added to config map
284	ETSTATUS_CONVERTING_BCD_TO_HEX	Error in converting BCD to hex
285	ETSTATUS_ERROR_ALREADY_EXISTS	The object already exists
286	ETSTATUS_NULL_HASH_MAP	Function called with null hash map object
287	ETSTATUS_HASH_MAP_NOT_INITIALIZED	Function called with uninitialized hash map
288	ETSTATUS_HASH_MAP_CORRUPTED	Detected hashmap corruption
289	ETSTATUS_JSON_PARSER_ERROR	JSON parsing failed
290	ETSTATUS_JSON_GET_ERROR	JSON query failed
291	ETSTATUS_NULL_JSON_OBJECT	Function called with null JSON object
292	ETSTATUS_JSON_NON_ARRAY_TYPE	Received non array type where array is expected
293	ETSTATUS_JAVA_GET_ENV_FAILED	Failed to get Java environment from JVM instance
294	ETSTATUS_INVALID_ENV	JAVA_HOME environment variable does not exist
295	ETSTATUS_INVALID_VALIDITY_INTERVAL	Validation Gateway (VG) response validity interval is incorrect
296	ETSTATUS_LOAD_CONFIG_ERROR	Failed to load config file from the provided config URL
297	ETSTATUS_INVALID_AUTH_CRED	Invalid authentication credentials (E401-77-03)

13.16. Plugin License & Device Errors (298-310)

Code	Name	Description
298	ETSTATUS_PLUGIN_LICENSE_INVALID	Plugin reported that the usage license is not valid

Code	Name	Description
299	ETSTATUS_PLUGIN_LICENSE_NOT_FOUND	Plugin reported that the usage license not found
300	ETSTATUS_PLUGIN_LICENSE_EXPIRED	Plugin reported that the usage license expired
301	ETSTATUS_PLUGIN_NOT_SUPPORTED	Plugin reported that the environment or device not supported
302	ETSTATUS_INVALID_DATA_PROT_KEY	Invalid data protection public key
303	ETSTATUS_NO_READER_WITH_EMIRATES_ID	No smartcard reader found with Emirates ID card
307	ETSTATUS_ID_VER_NOT_SUPPORTED	Emirates ID version is not supported
308	ETSTATUS_PLUGIN_DIR_PATH_ERROR	Failed to get plugin directory path from config file
309	ETSTATUS_AGENT_TKT_VER_MISMATCH	Agent and Toolkit version mismatch
310	ETSTATUS_FP_CLOSE_DEV_ERROR	Failed to disconnect fingerprint sensor
311	ETSTATUS_INVALID_COMBI_READER_CONFIG	Invalid or incomplete combi reader plugin configuration
312	ETSTATUS_DUP_PLG_LIST_READERS	Plugin has multiple list reader functions
313	ETSTATUS_VG_RESP_CERT_INVALID	Validation Gateway (VG) response signing certificate is invalid
314	ETSTATUS_LIC_LIMIT_REACHED	Failed to parse "config_li" file as the number of services in the Toolkit license exceeds the limit
315	ETSTATUS_INVALID_TEMP_TAG	Invalid template tag in the ID Card raw data
316	ETSTATUS_AUTH_CERT_NOT_FOUND	Failed to retrieve authentication certificate from the PKI certificate file
317	ETSTATUS_NFC_NOT_SUPPORTED	Plugin does not support NFC capability
318	ETSTATUS_NFC_TAG_NOT_SET	NFC tag is not set
319	ETSTATUS_INVALID_MRZ_DATA	Invalid MRZ string

Code	Name	Description
320	ETSTATUS_INVALID_MRZ_CHECKSUM	MRZ string contains invalid checksum
321	ETSTATUS_SVC_NOT_RUNNING	Plugin service is not installed/running
322	ETSTATUS_ID_SCHEMA_CORRUPTED	ID card schema got corrupted
323	ETSTATUS_ID_CMD_LIMIT_EXCEEDED	ID card command limit reached
324	ETSTATUS_INVALID_CONTENT_LEN_ERROR	HTTP response header received has "content-length" zero
325	ETSTATUS_NO_CONTENT_LENGTH	HTTP response header does not contain "content-length" attribute
326	ETSTATUS_URL_BAD_FORMAT	Provided URL is of invalid format
327	ETSTATUS_NO_SIG_DATA	Failed to retrieve detached signature data

13.17. Card Container & Data File Errors (329-345)

Code	Name	Description
329	ETSTATUS_DATA_FILE_NOT_FOUND	Data file not found
330	ETSTATUS_READ_AC_LOCKED	Read access condition locked
331	ETSTATUS_WRITE_AC_LOCKED	Update access condition locked
332	ETSTATUS_KEY_FILE_NOT_FOUND	Key file not found
333	ETSTATUS_PIN_FILE_NOT_FOUND	PIN file not found
334	ETSTATUS_OBJ_LIMIT_REACHED	Maximum object limit reached
335	ETSTATUS_INVALID_ENC_TYPE	Invalid encoding type
336	ETSTATUS_CARD_NOT_PERS	Emirates ID Card is not personalized
337	ETSTATUS_ATTR_LIMIT_REACHED	Maximum attribute limit reached
338	ETSTATUS_INVALID_CMD_COUNT	Invalid data file command count
339	ETSTATUS_INVALID_DF_TYPE	Invalid dedicated file type
340	ETSTATUS_CONT_SCH_NOT_LOADED	Failed to load container schema
341	ETSTATUS_INVALID_CONTAINER_TYPE	Invalid container type
342	ETSTATUS_INVALID_FILE_TYPE	Invalid file type

Code	Name	Description
343	ETSTATUS_CMD_LIMIT_REACHED	Number of commands in the response exceeds the limit
344	ETSTATUS_INVALID_AC	Data file is created using invalid access conditions
345	ETSTATUS_ID_CONT_ATTR_LIMIT_EXCEEDED	Emirates ID card container attribute type limit reached

13.18. Miscellaneous Errors (346-366)

Code	Name	Description
346	ETSTATUS_ERROR_OP_TIMEOUT	Operation timed out
347	ETSTATUS_LOAD_MOD_HANDLE_ERROR	Failed to load module handle
348	ETSTATUS_NO_SET_NFC_TAG_METHOD	Loaded plugins does not contain method to set NFC tag object
349	ETSTATUS_NODES_LIMIT_REACHED	Number of allowed nodes in the license exceeds the limit
350	ETSTATUS_SC_READER_SN_ERROR	Error in retrieving serial number from connected smart card reader
351	ETSTATUS_ESB_SVC_NAME_ERROR	Failed to get ESB service name
352	ETSTATUS_ESB_SRC_CHANNEL_ERROR	Failed to get ESB source channel
353	ETSTATUS_ESB_SVC_VER_ERROR	Failed to get ESB service version
354	ETSTATUS_ESB_SVC_LANG_ERROR	Failed to get ESB service language
355	ETSTATUS_ESB_USER_NAME_ERROR	Failed to get ESB user name
356	ETSTATUS_ESB_PWD_ERROR	Failed to get ESB password
357	ETSTATUS_ESB_KEY_ERROR	Failed to get ESB shared key
358	ETSTATUS_ESB_CONSUMER_ID_ERROR	Failed to get ESB consumer ID
359	ETSTATUS_NO_SESSION_ID	Response header does not contain session ID (Set-Cookie)
360	ETSTATUS_ESB_ERROR	Received error response from ESB interface
361	ETSTATUS_INVALID_PUB_KEY	Data signature validation failed as the key used for validation is not matching with the signing key

Code	Name	Description
362	ETSTATUS_HASH_VALIDATION_ERROR	Failed to validate hash of the signature data
363	ETSTATUS_CONFIG_LOC_NOT_PROVIDED	Config URL and directory not provided
364	ETSTATUS_PARSE_APP_CONFIG_ERROR	Failed to parse app configuration from config file
365	ETSTATUS_PARSE_CONFIG_PARAMS_ERROR	Failed to parse config params
366	ETSTATUS_PARSE_EF_DATA_ERROR	Failed to parse EF data

14. Face Authentication Parameters

Face authentication supports ID-based and passport-based authentication with the following liveness detection modes:

Parameter	Description
PASSIVE_LIVENESS	Automatic liveness detection without user interaction
SMILE_LIVENESS	Requires the user to smile for liveness confirmation
MAGNIFY_EYE_LIVENESS	Uses eye magnification for liveness detection

Face authentication functions (deprecated):

- `AuthenticateFaceOnServerWithID` (deprecated) – authenticate using Emirates ID number
- `AuthenticateFaceOnServerWithPassport` (deprecated) – authenticate using passport number
- `AuthenticateFaceOnServerWithIDEx` (deprecated) – extended face authentication with ID, same as `AuthenticateFaceOnServerWithID` with additional parameters
- `AuthenticateFaceOnServerWithPassportEx` (deprecated) – extended face authentication with passport

Note: Connection to the ID Card is not required for face authentication. Face authentication can be performed with or without a card connection.

15. Additional API Functions

The following functions are available in the Toolkit API in addition to the core services documented in the SDK developer guides.

15.1. Extended Card Connection

- `ConnectEx` – extended card connect with additional parameters
- `DisconnectEx` – extended card disconnect
- `ReadPublicDataEx` – extended public data read
- `CleanupEFDataMemory` – frees memory allocated by `ReadPublicDataEF`

15.2. NFC

- `SetNFCTag` – sets the NFC tag handle for NFC operations

15.3. Development and Testing

- `SetTestProvider` – sets a test provider for development and testing